

## B.2 EVALUATION IMPLEMENTATION DETAILS

All evaluations were implemented on 3 NVIDIA A100-40G with VLMEvalKit v0.2. Following the idea of Duan et al. (2025), all the models used very low temperatures or temperatures equal to 0 and set `do_sample = False` to ensure reproducibility and certainty of the results. The API checkpoints for proprietary models are listed in Table 12.

**Table 12:** Proprietary APIs evaluated in this paper

| Proprietary Model & provider           | API endpoint                          |
|----------------------------------------|---------------------------------------|
| Claude 3.7 Sonnet (Anthropic)          | claude-3-7-sonnet-20250610            |
| Doubao 1.5 VL (volcengine)             | doubao-1-5-vision-pro-32k-250115      |
| Doubao 1.5 VL-thinking (volcengine)    | doubao-1-5-thinking-vision-pro-250428 |
| Gemini 2.0 Flash (Google)              | gemini-2.0-flash                      |
| Gemini 2.5 Flash (Google)              | gemini-2.5-flash-preview-05-20        |
| Gemini 2.5 Flash w/o thinking (Google) | gemini-2.5-flash-preview-05-20        |
| GPT-4.1 nano (OpenAI)                  | gpt-4.1-nano-2025-04-14               |
| GPT-4o (OpenAI)                        | gpt-4o-2024-08-06                     |
| GPT-4o-mini (OpenAI)                   | gpt-4o-mini-2025-06-10                |
| GPT-5 (OpenAI)                         | gpt-5                                 |
| o4-mini (OpenAI)                       | o4-mini-2025-04-16                    |

The prompt templates used in the evaluation for different models are shown below:

**Listing 1:** Prompt Templates used for Proprietary Models in VLMEvalKit

```
PROMPT_TEMPLATES = {
    "SYSTEM": "You are a helpful assistant.",

    "USER": """<image>
        Question: The two images above show a 3D structure from
        different angles. Which one of the options below could be
        constructed to appear the same as both given views when observed
        from the corresponding perspectives without rotation and overlaps?
        Select the most likely one.
        Options:
        A. A
        B. B
        C. C
        D. D
        Answer with the option's letter from the given choices
        directly."""
}
```

**Listing 2:** Prompt Templates used for Llama Serie Models in VLMEvalKit

```
PROMPT_TEMPLATES = {
    "SYSTEM": "",

    "USER": """<|begin_of_text|><|start_header_id|>user<|end_header_id|>

<im_start><image><im_end>
        Question: The two images above show a 3D structure from
        different angles. Which one of the options below could be
        constructed to appear the same as both given views when observed
        from the corresponding perspectives without rotation and overlaps?
        Select the most likely one.
        Options:
        A. A
```

```

        B. B
        C. C
        D. D
    Answer with the option's letter from the given choices
    directly.<|eot_id|>""
}

```

**Listing 3:** Prompt Templates used for QwenVL, InternVL, Ovis2 Serie Models in VLMEvalKit

```

PROMPT_TEMPLATES = {
    "USER": ""<image>
        Question: The two images above show a 3D structure from
        different angles. Which one of the options below could be
        constructed to appear the same as both given views when observed
        from the corresponding perspectives without rotation and overlaps?
        Select the most likely one.
        Options:
        A. A
        B. B
        C. C
        D. D
        Please select the correct answer from the options above.""
}

```

**Listing 4:** Prompt Templates used for VLM-R1 and LMM-R1 in VLMEvalKit

```

PROMPT_TEMPLATES = {
    "USER": ""<image>
        Question: The two images above show a 3D structure from
        different angles. Which one of the options below could be
        constructed to appear the same as both given views when observed
        from the corresponding perspectives without rotation and overlaps?
        Select the most likely one.
        Options:
        A. A
        B. B
        C. C
        D. D
        Please select the correct answer from the options above. Output
        the thinking process in <think> </think> and final answer in
        <answer> </answer> tags.""
}

```

**Listing 5:** Prompt Templates used for VLAA\_Thinker Serie Models in VLMEvalKit

```

PROMPT_TEMPLATES = {
    "SYSTEM": "You are VL-Thinking, a helpful assistant with excellent
    reasoning ability. You should first think about the reasoning
    process and then provide the answer. Use <think>...</think> and
    <answer>...</answer> tags."

    "USER": ""<image>
        Question: The two images above show a 3D structure from
        different angles. Which one of the options below could be
        constructed to appear the same as both given views when observed
        from the corresponding perspectives without rotation and overlaps?
        Select the most likely one.
        Options:
        A. A
        B. B
        C. C
        D. D
        Please select the correct answer from the options above.""
}

```

### B.3 MORE EVALUATION RESULTS

**Table 13:** Different-task accuracies on Spatial-DISE Bench. Abbreviations—2D Comb.: 2D Combination; 2D Fold.: 2D Folding; 2D Rot.: 2D Rotation; 2D S.F.: 2D Shape Finding; 3D Comb.: 3D Combination; 3D Fold.: 3D Folding; 3D Proj.: 3D Projection; 3D Rot.: 3D Rotation; 3D S.F.: 3D Shape Finding; F&P: Fold and Punch. **Bold** indicates the highest accuracy; Underline indicates the second highest.

| Model                                           | Acc.         | 2D Comb.     | 2D Fold.     | 2D Rot.      | 2D S.F.      | 3D Comb.     | 3D Fold.     | 3D Proj.     | 3D Rot.      | 3D S.F.      | F&P          |
|-------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Proprietary</b>                              |              |              |              |              |              |              |              |              |              |              |              |
| Claude 3.7 Sonnet                               | 30.6%        | <u>29.2%</u> | 25.6%        | 30.4%        | <b>38.1%</b> | 20.0%        | <u>50.7%</u> | 31.4%        | 31.4%        | <u>28.8%</u> | 24.4%        |
| Doubao1.5 VL                                    | 33.8%        | <b>41.7%</b> | 25.6%        | <b>43.5%</b> | 33.3%        | <u>26.7%</u> | <u>44.9%</u> | <b>37.1%</b> | <u>37.1%</u> | <b>34.8%</b> | 25.6%        |
| Gemini 2.0 Flash                                | <b>34.2%</b> | 20.8%        | <b>41.0%</b> | 30.4%        | 23.8%        | <u>26.7%</u> | <b>56.5%</b> | 31.4%        | <b>51.4%</b> | 15.2%        | 24.4%        |
| GPT4.1 nano                                     | 29.3%        | 29.2%        | 35.9%        | 30.4%        | 14.3%        | <b>30.0%</b> | 36.2%        | 35.7%        | 30.0%        | 18.2%        | 23.1%        |
| GPT4o                                           | 28.1%        | <u>29.2%</u> | 26.9%        | 17.4%        | 33.3%        | 25.0%        | 30.4%        | <u>22.9%</u> | 32.9%        | 25.8%        | <b>33.3%</b> |
| GPT4o-mini                                      | 25.6%        | 20.8%        | 28.2%        | 30.4%        | 28.6%        | 15.0%        | 37.7%        | 21.4%        | 22.9%        | 28.8%        | 23.1%        |
| Gemini 2.5 Flash                                | 31.5%        | 12.5%        | 33.3%        | 17.4%        | 33.3%        | 18.3%        | 69.6%        | 27.1%        | 40.0%        | 16.7%        | 24.4%        |
| Gemini 2.5 Flash w/o thinking                   | 32.0%        | 12.5%        | 33.3%        | 17.4%        | 33.3%        | 16.7%        | 69.6%        | 28.6%        | 40.0%        | 19.7%        | 25.6%        |
| GPT-5                                           | 30.1%        | 20.8%        | 30.8%        | 43.5%        | 14.3%        | 25.0%        | 31.9%        | 25.7%        | 45.7%        | 30.3%        | 24.4%        |
| o4-mini                                         | 33.3%        | 33.3%        | 30.8%        | 52.2%        | 23.8%        | 10.0%        | 47.8%        | 25.7%        | 38.6%        | 48.5%        | 26.9%        |
| Proprietary Average                             | <u>30.9%</u> | <u>25.0%</u> | <u>31.1%</u> | <u>31.3%</u> | <u>27.6%</u> | <u>21.3%</u> | <u>47.5%</u> | <u>28.7%</u> | <u>37.0%</u> | <u>26.7%</u> | <u>25.5%</u> |
| <b>Open-source</b>                              |              |              |              |              |              |              |              |              |              |              |              |
| Llama-3V-11B                                    | 24.5%        | 29.2%        | 24.4%        | 21.7%        | 19.0%        | 30.0%        | 31.9%        | 14.3%        | 24.3%        | 25.8%        | 23.1%        |
| Cambrian-13b                                    | 26.7%        | 20.8%        | <u>30.8%</u> | 30.4%        | 23.8%        | <u>26.7%</u> | 21.7%        | <b>32.9%</b> | 25.7%        | <u>27.3%</u> | 23.1%        |
| Cambrian-8b                                     | 22.9%        | 25.0%        | 26.9%        | 30.4%        | <b>33.3%</b> | 16.7%        | 33.3%        | 15.7%        | 15.7%        | <u>27.3%</u> | 17.9%        |
| InternVL3-38B                                   | <b>32.4%</b> | <u>29.2%</u> | 28.2%        | <b>47.8%</b> | 23.8%        | 26.7%        | 42.0%        | 30.0%        | 40.0%        | <u>27.3%</u> | 30.8%        |
| InternVL3-14B                                   | 31.1%        | 25.0%        | 24.4%        | 21.7%        | 14.3%        | 20.0%        | <b>53.6%</b> | 31.4%        | <b>42.9%</b> | 19.7%        | <b>34.6%</b> |
| InternVL3-8B                                    | 26.3%        | <b>33.3%</b> | <b>35.9%</b> | 30.4%        | 14.3%        | 20.0%        | 29.0%        | 28.6%        | 32.9%        | 9.1%         | 25.6%        |
| Kimi-VL-A3B                                     | 24.3%        | 12.5%        | 29.5%        | 26.1%        | <b>33.3%</b> | 20.0%        | 26.1%        | 27.1%        | 35.7%        | 10.6%        | 20.5%        |
| Ovis2-16B                                       | 26.3%        | 20.8%        | 16.7%        | 13.0%        | 19.0%        | 20.0%        | 52.2%        | 27.1%        | <b>42.9%</b> | 10.6%        | 23.1%        |
| Ovis2-8B                                        | 23.8%        | 25.0%        | 28.2%        | 17.4%        | 28.6%        | 11.7%        | 36.2%        | 21.4%        | 34.3%        | 7.6%         | 24.4%        |
| Qwen2.5-VL-32B                                  | 27.2%        | 20.8%        | 19.2%        | 21.7%        | <u>23.8%</u> | 21.7%        | 34.8%        | 31.4%        | 35.7%        | <b>28.8%</b> | 24.4%        |
| Qwen2.5-VL-7B                                   | 26.1%        | <b>33.3%</b> | 26.9%        | <u>39.1%</u> | <b>33.3%</b> | <b>31.7%</b> | 30.4%        | <u>24.3%</u> | 32.9%        | 10.6%        | 17.9%        |
| Qwen2.5-VL-3B                                   | 22.9%        | <u>29.2%</u> | 28.2%        | 17.4%        | 14.3%        | 23.3%        | 36.2%        | 17.1%        | 22.9%        | 12.1%        | 21.8%        |
| Open-source Average                             | <u>26.2%</u> | <u>25.3%</u> | <u>26.6%</u> | <u>26.4%</u> | <u>23.4%</u> | <u>22.4%</u> | <u>35.6%</u> | <u>25.1%</u> | <u>32.2%</u> | <u>18.1%</u> | <u>23.9%</u> |
| <b>Reasoning &amp; Spatial-Specified Models</b> |              |              |              |              |              |              |              |              |              |              |              |
| LLaVA-CoT                                       | 24.0%        | 29.2%        | <b>34.6%</b> | 13.0%        | 9.5%         | 30.0%        | 17.4%        | 22.9%        | 22.9%        | 19.7%        | 25.6%        |
| LMM-R1                                          | 26.1%        | 29.2%        | 28.2%        | 21.7%        | 38.1%        | 30.0%        | 36.2%        | 20.0%        | 24.3%        | 22.7%        | 19.2%        |
| VLM-R1                                          | 30.8%        | 25.0%        | 26.9%        | <u>39.1%</u> | <u>38.1%</u> | 36.7%        | 47.8%        | 18.6%        | 30.0%        | <u>24.2%</u> | 29.5%        |
| Kimi-VL-A3B-Thinking                            | 24.7%        | 16.7%        | 26.9%        | 26.1%        | <b>42.9%</b> | 31.7%        | 26.1%        | 28.6%        | 22.9%        | 15.2%        | 19.2%        |
| Doubao1.5-VL-thinking                           | <b>42.0%</b> | <b>62.5%</b> | 28.2%        | <b>43.5%</b> | 23.8%        | <b>61.7%</b> | <b>56.5%</b> | <b>31.4%</b> | <b>50.0%</b> | <b>39.4%</b> | 30.8%        |
| VLAA-Thinker-3B                                 | 25.9%        | 37.5%        | 20.5%        | 26.1%        | 28.6%        | 25.0%        | 36.2%        | 30.0%        | 27.1%        | 9.1%         | 28.2%        |
| VLAA-Thinker-7B                                 | 27.9%        | 25.0%        | 25.6%        | 26.1%        | 38.1%        | 28.3%        | 31.9%        | 27.1%        | 35.7%        | 22.7%        | 23.1%        |
| SpaceThinker                                    | 32.6%        | 29.2%        | 20.5%        | <b>43.5%</b> | 33.3%        | 43.3%        | 49.3%        | 22.9%        | 35.7%        | 22.7%        | 33.3%        |
| SpaceOm                                         | <u>25.9%</u> | 25.0%        | 14.1%        | <b>43.5%</b> | 33.3%        | 36.7%        | 49.3%        | 24.3%        | 32.9%        | 24.2%        | <b>37.2%</b> |
| SpaceR                                          | 27.0%        | 37.5%        | 32.1%        | 34.8%        | 28.6%        | 26.7%        | 29.0%        | 17.1%        | 37.1%        | 21.2%        | 19.2%        |
| Reasoning & Spatial Average                     | 27.6%        | 27.8%        | 25.9%        | 28.6%        | 27.0%        | 28.2%        | 37.1%        | 25.1%        | 31.8%        | 19.8%        | 25.4%        |
| Overall Average                                 | 28.4%        | 27.2%        | 27.8%        | 29.6%        | 27.2%        | 26.0%        | <u>40.1%</u> | 26.0%        | 33.6%        | 22.0%        | 25.2%        |
| SpaceOm-sft                                     | 33.8%        | 25.0%        | 25.6%        | 26.1%        | 23.8%        | 45.0%        | 46.4%        | 31.4%        | 50.0%        | 30.3%        | 20.5%        |
| Qwen2.5-VL-7B-sft                               | 47.0%        | 33.3%        | 34.6%        | 34.8%        | 9.5%         | 78.3%        | 69.6%        | 41.4%        | 65.7%        | 69.7%        | 21.8%        |
| Human                                           | 76.8%        | 75.2%        | 76.5%        | 78.1%        | 61.5%        | 55.4%        | 86.6%        | 81.1%        | 82.0%        | 81.8%        | 76.8%        |

### B.4 SUPERVISED FINE-TUNING HYPERPARAMETERS

The Supervised Fine-Tuning (SFT) experiments were conducted using the Swift framework. We employed the Low-Rank Adaptation (LoRA) technique to efficiently fine-tune both the Qwen2.5-VL-7B and SpaceOm models on the Spatial-DISE-12K training set. All linear layers of the models were targeted for LoRA adaptation. The key hyperparameters used for the fine-tuning process are detailed in Table 13.

**Table 14:** Hyperparameters for SFT Training

| Hyperparameter          | Value      |
|-------------------------|------------|
| Framework               | Swift      |
| Fine-Tuning Method      | LoRA       |
| Target Modules          | all-linear |
| LoRA Rank (lora_rank)   | 8          |
| LoRA Alpha (lora_alpha) | 32         |
| Batch Size              | 48         |
| Precision (torch_dtype) | bfloat16   |
| Learning Rate           | 1.5e-4     |
| Warmup Ratio            | 0.05       |
| Number of Epochs        | 2          |
| Max Sequence Length     | 4096       |
| Deepspeed               | zero3      |